Now, $\sin 4B = 2\sin 2B \cos 2B$

$$= 2\frac{2\tan B}{1+\tan^2 B} \times \frac{1-\tan^2 B}{1+\tan^2 B}$$

$$= \frac{4\times\frac{1}{3}}{1+\frac{1}{9}} \times \frac{1-\frac{1}{9}}{1+\frac{1}{9}} = \frac{24}{25} \qquad \ldots (2)$$

From(1) and (2) we get, $\cos 2A = \sin 4B$.

Example 4.12

If $\tan A = \frac{1-\cos B}{\sin B}$ then prove that $\tan 2A = \tan B$

Solution

Consider $\frac{1-\cos B}{\sin B} = \frac{2\sin^2 \frac{B}{2}}{2\sin\frac{B}{2}\cos\frac{B}{2}}$

$$= \frac{\sin\frac{B}{2}}{\cos\frac{B}{2}} = \tan\frac{B}{2}$$

$$\tan A = \frac{1-\cos B}{\sin B}$$

$$\tan A = \tan\frac{B}{2}$$

$$A = \frac{B}{2}$$

$$2A = B$$

$$\therefore \tan 2A = \tan B$$

Example 4.13

If $\tan\alpha = \frac{1}{3}$ and $\tan\beta = \frac{1}{7}$ then prove that $(2\alpha+\beta) = \frac{\pi}{4}$.

Solution

$$\tan(2\alpha+\beta) = \frac{\tan 2\alpha + \tan\beta}{1-\tan 2\alpha \cdot \tan\beta}$$

$$\tan 2\alpha = \frac{2\tan\alpha}{1-\tan^2\alpha} = \frac{2\times\frac{1}{3}}{1-\frac{1}{9}} = \frac{3}{4}$$

$$\tan(2\alpha+\beta) = \frac{\frac{3}{4}+\frac{1}{7}}{1-\frac{3}{4}\times\frac{1}{7}} = \frac{\frac{25}{28}}{\frac{25}{28}}$$

$$= 1 = \tan\frac{\pi}{4}$$

$$2\alpha+\beta = \frac{\pi}{4}$$

Exercise 4.2

1. Find the values of the following:
   (i) cosec15° (ii) sin(–105°) (iii) cot75°
2. Find the values of the following
   (i) sin76° cos16° + cos76° sin16°
   (ii) $\sin\frac{\pi}{4}\cos\frac{\pi}{12} + \cos\frac{\pi}{4}\sin\frac{\pi}{12}$
   (iii) cos70° cos10° – sin70° sin10°
   (iv) $\cos^2 15° - \sin^2 15°$
3. If $\sin A = \frac{3}{5}, 0 < A < \frac{\pi}{2}$ and $\cos B = \frac{-12}{13}$, $\pi < B < \frac{3\pi}{2}$ find the values of the following:
   (i) cos(A+B) (ii) sin(A–B)
   (iii) tan(A–B)
4. If $\cos A = \frac{13}{14}$ and $\cos B = \frac{1}{7}$ where A, B are acute angles prove that A–B = $\frac{\pi}{3}$
5. Prove that 2tan80° = tan85° –tan 5°
6. If $\cot\alpha = \frac{1}{2}$, $\sec\beta = \frac{-5}{3}$, where $\pi < \alpha < \frac{3\pi}{2}$ and $\frac{\pi}{2} < \beta < \pi$, find the value of $\tan(\alpha+\beta)$. State the quadrant in which $\alpha+\beta$ terminates.
7. If A+B = 45°, prove that (1+tanA)(1+tanB) = 2 and hence deduce the value of $\tan 22\frac{1^\circ}{2}$
8. Prove that
   (i) $\sin(A+60^\circ)+\sin(A-60^\circ) = \sin A$
   (ii) $\tan 4A \tan 3A \tan A + \tan 3A + \tan A - \tan 4A = 0$
9. (i) If $\tan\theta = 3$ find $\tan 3\theta$
   (ii) If $\sin A = \frac{12}{13}$, find $\sin 3A$
10. If $\sin A = \frac{3}{5}$, find the values of cos3A and tan3A
11. Prove that $\frac{\sin(B-C)}{\cos B\cos C} + \frac{\sin(C-A)}{\cos C\cos A} + \frac{\sin(A-B)}{\cos A\cos B} = 0$.

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